Using Fisher Information to Evaluate the Preservation of Spatial Discriminability in HRTF Upsampling

Dominic Bassett, Matthew Cheshire, and Samuel Smith

Abstract—**HRTF upsampling is typically evaluated using signal-domain reconstruction errors and localisation-model summaries, neither of which directly measures preservation of local spatial discriminability. We introduce a Fisher information evaluation defined on the sphere of source directions. Spatial gradients of positive spectral-gradient, interaural level-difference, and interaural time-difference features form a bivariate Fisher information tensor, whose inverse gives the Cramér–Rao bound on the covariance of an unbiased local angular estimator. Reference and reconstructed tensors are compared using the affine-invariant Riemannian metric, measuring changes in the scale, anisotropy, and orientation of local spatial discrimination ellipses. We evaluate the method on a 41-subject SONICOM cohort under four sparse-grid retention conditions. The comparison includes SUpDEq-based interpolation, RANF, and FSP-AE, and reports Fisher-tensor error alongside log-spectral distance and Bayesian localisation metrics. RANF achieves the lowest Fisher-tensor and signal-domain errors, while FSP-AE maintains stable tensor errors despite higher log-spectral distance. Projected $d' = 1$ thresholds express selected tensors in angular units and show the expected increase in lateral threshold from frontal to lateral directions. The proposed evaluation therefore quantifies reconstruction errors in local spatial discriminability that are not measured by signal-fidelity or localisation summaries alone.**

Index Terms—**Fisher information, head-related transfer functions, information geometry, spatial audio, spatial discriminability, upsampling.**

I. Introduction and Background

Head-related transfer functions (HRTFs) encode the direction-dependent filtering imposed by the listener's head, torso, and pinnae. Dense individual HRTF measurements are expensive to obtain, so spatial upsampling may be used to reconstruct a dense directional field from a sparse set of measured directions. Upsampling methods are usually evaluated with log-spectral distance (LSD), interaural level-difference error, interaural time-difference error, and localisation-model summaries. These quantities measure signal fidelity or predicted pointing performance, but they do not directly ask whether the reconstructed field preserves the listener's local ability to discriminate neighbouring directions.

The relevant perceptual object is therefore not only the HRTF at a single direction, but the local variation of localisation cues over the sphere. Spatial discrimination studies often report lateral and polar acuity separately, as minimum audible angles or response variability along azimuthal and elevational dimensions. An early attempt to combine these dimensions was made by Starch [3], who represented horizontal and vertical discrimination measurements as "auditory discrimination ellipses". His construction is historically useful because it recognises that spatial acuity is bivariate and direction dependent. The construction differs from the present one in two important respects: the ellipses were assembled from separate horizontal and vertical measurements as axis-aligned summaries, rather than from a coupled local tensor, and the measurements used a 100-Hz electric-fork stimulus rather than broadband HRTF cues. They therefore did not include the cross-band spectral structure normally involved in elevation discrimination.

Modern localisation data support the need for a coupled bivariate account. Carlile et al. [6] modelled localisation-error distributions using Fisher or Kent distributions. In the pooled data, 50 of 76 target locations (66%) were Kent distributed. Their major axes often departed from the azimuth/elevation axes and, rather than following the cones of confusion expected under an orthogonal cue-processing account, showed co-latitude alignment laterally and posteriorly and radial alignment near the frontal midline. The Bayesian model of Barumerli et al. [9] is calibrated against aggregate lateral, polar, and quadrant errors and provides realistic point estimates. However, its post-decision isotropic von Mises–Fisher scatter causes Monte Carlo response distributions to converge towards radial symmetry about the MAP estimate (Fig. 2). After repeated simulation, the response variance therefore mainly reflects the imposed response-scatter stage rather than the anisotropic directional precision reported by Carlile et al. These observations are directly relevant to HRTF upsampling because a reconstruction may preserve average spectral content while altering the scale, anisotropy, or orientation of local discrimination. A tensorial sensory analysis is therefore needed alongside localisation-model evaluation.

Fisher information gives such a local tensor: it quantifies how rapidly the expected auditory cues change with small displacements in source direction. Within auditory neuroscience, Fisher information has been used to relate binaural cue statistics and neural response properties. Harper and McAlpine [7] analysed the population coding of interaural time differences (ITDs) in the mammalian brainstem and showed that neural tuning distributions can be organised to maximise Fisher information over naturally occurring cue ranges. Pavão et al. [8] extended this analysis to human listeners and reported that the spatial distribution of Fisher information derived from HRTF statistics correlates with measured free-field ITD

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discrimination thresholds.

This paper uses Fisher information to evaluate HRTF reconstruction and spatial upsampling. Let the sound-source direction be $\mathbf{r} \in \mathbb{S}^2$, and let $\boldsymbol{\xi} \in \mathbb{R}^2$ denote local angular coordinates in an orthonormal tangent basis at $\mathbf{r}$. The Fisher information matrix quantifies local stimulus discriminability on these coordinates. For any unbiased local estimator $\hat{\boldsymbol{\xi}}$, provided that the Fisher information matrix is nonsingular, the Cramér–Rao bound (CRB) states

$$\mathrm{Cov}(\hat{\boldsymbol{\xi}}) \succeq \mathbf{g}(\mathbf{r})^{-1}, \tag{1}$$

where $\mathbf{g}(\mathbf{r}) \in \mathbb{R}^{2\times 2}$ is the Fisher information matrix evaluated on the local tangent plane. Thus, under the assumed cue model, $\mathbf{g}(\mathbf{r})^{-1}$ is the Cramér–Rao bound on the covariance of any unbiased local angular estimator [1].

The FIM defines a Riemannian metric on the parameter manifold [2]. Its squared infinitesimal distance is

$$ds^2 = d\boldsymbol{\xi}^T \mathbf{g}(\mathbf{r}) d\boldsymbol{\xi}. \tag{2}$$

For the cue vector $\mathbf{q}$, assume $\mathbf{q} \mid \boldsymbol{\xi} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\xi}), \boldsymbol{\Sigma})$, with direction-independent internal-noise covariance $\boldsymbol{\Sigma}$. For two nearby directions separated by $\delta\boldsymbol{\xi}$, equal-covariance Gaussian signal detection theory gives $d'^2 = \Delta\boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \Delta\boldsymbol{\mu}$. Linearising the cue mean as $\Delta\boldsymbol{\mu} \approx \mathbf{J}\delta\boldsymbol{\xi}$ yields

$$d'^2 \approx \delta\boldsymbol{\xi}^T \underbrace{\mathbf{J}^T \boldsymbol{\Sigma}^{-1} \mathbf{J}}_{\mathbf{g}(\mathbf{r})} \delta\boldsymbol{\xi}. \tag{3}$$

Writing $\delta\boldsymbol{\xi} = \rho\mathbf{u}$, where $\mathbf{u} \in \mathbb{R}^2$ is a unit tangent direction, gives the angular displacement predicted to reach $d' = 1$:

$$\rho_{d'=1}(\mathbf{r}, \mathbf{u}) = \left(\mathbf{u}^T \mathbf{g}(\mathbf{r}) \mathbf{u}\right)^{-1/2}. \tag{4}$$

This is the scalar threshold reported later for selected lateral and vertical directions. The CRB statement above gives the covariance bound $\mathbf{g}(\mathbf{r})^{-1}$; its eigenvectors are the same as those of $\mathbf{g}(\mathbf{r})$, with reciprocal eigenvalues. Thus the principal axes of the $d' = 1$ contour are also the principal axes of the CRB covariance ellipse. The resulting ellipse is the analytical discrimination ellipse implied by the observer model, rather than a concatenation of separately measured lateral and polar thresholds; Fig. 1 shows the corresponding SONICOM ellipses.

The cue model uses positive spectral-gradient, interaural level-difference, and transformed interaural time-difference features following the sensory evidence used by Barumerli et al. [9]. The cue-family noise covariance is diagonal, so monaural spectral, ILD, and ITD noise are treated as independent at the feature level. This does not make the spatial tensor diagonal. For diagonal $\boldsymbol{\Sigma}$, the off-diagonal tensor entry is $g_{12} = \sum_i \sigma_i^{-2}(\partial_1\mu_i)(\partial_2\mu_i)$. It is generally nonzero because cue gradients need not align with the tangent axes. Setting $g_{12} = 0$ would force the discrimination ellipses to remain axis-aligned and remove cross-dimensional sensitivity between lateral and polar displacements. The analysis therefore retains the full tensor.

The evaluation compares the Fisher tensors of measured and reconstructed HRTFs together with signal-domain errors and Bayesian observer simulations. Fisher information measures how the localisation cues vary between neighbouring directions, so reconstructions with similar spectral error can produce different Fisher tensors.

Reference and reconstructed Fisher tensors are compared using the affine-invariant Riemannian metric (AIRM) on symmetric positive-definite matrices. The resulting discrepancy measures differences in the size, shape, and orientation of local discrimination ellipses. It tests whether an upsampling method preserves the local discriminability represented by the measured Fisher tensor field.

II. Cue Feature Extraction

Let the source direction be represented by a unit Cartesian vector

$$\mathbf{r} = [r_x, r_y, r_z]^T \in \mathbb{S}^2.$$

For ear $c \in \{L, R\}$, let $A_c(f_k, \mathbf{r})$ and $\widehat{A}_c(f_k, \mathbf{r})$ denote the target and reconstructed HRTF magnitudes at frequency bin f_k, $k \in \{1, \ldots, F\}$. Let $D_c(f_k, \mathbf{r})$ and $\widehat{D}_c(f_k, \mathbf{r})$ denote their corresponding directional transfer function (DTF) magnitudes after applying the same common-transfer-function (CTF) removal convention. DTF cues are used for the Fisher analysis; raw HRTF magnitudes are retained for signal-domain error metrics.

Three cue functions define the Fisher analysis. They are extracted at measured directions and represented by smooth spherical-harmonic fields, whose spatial derivatives define the tensor.

A. Positive Spectral-Gradient Features (PGE)

Positive spectral-gradient features follow the sagittal-plane cue convention of Baumgartner et al. [14] as implemented in AMT `barumerli2023_featureextraction`. DTF magnitudes are interpolated onto 27 ERB-rate-spaced centre frequencies between 0.7 and 18 kHz, converted to dB, and differenced across an approximately one-ERB tonotopic interval. For ear $c \in \{L, R\}$,

$$S_c[b, \mathbf{r}] = \max(m_c[b+d, \mathbf{r}] - m_c[b, \mathbf{r}], 0), \tag{5}$$

where $m_c[b, \mathbf{r}] = 20\log_{10}(D_c(f_b, \mathbf{r}) + \varepsilon_A)$, $b = 1, \ldots, B-d$, and d is selected from the ERB bandwidth model to approximate a one-ERB comparison. The retained positive differences form the monaural spectral cue vector used by the Fisher analysis.

B. Interaural Level Difference (ILD)

The broadband ILD is defined as the log-ratio of average spectral power across the two ears:

$$\mathrm{ILD}(\mathbf{r}) = 10\log_{10}\left(\frac{\frac{1}{F}\sum_{k=1}^{F} D_L(f_k, \mathbf{r})^2 + \varepsilon_P}{\frac{1}{F}\sum_{k=1}^{F} D_R(f_k, \mathbf{r})^2 + \varepsilon_P}\right). \tag{6}$$

Here $\varepsilon_P > 0$ is the power-domain numerical floor used for both target and reconstructed cues.

C. Interaural Time Difference (ITD)

Let $a_L(t, \mathbf{r})$ and $a_R(t, \mathbf{r})$ denote the left- and right-ear HRIRs at direction $\mathbf{r}$. For HRTF sets represented as HRIRs, or as complex responses from which HRIRs are obtained by inverse transformation, the broadband ITD field is estimated from the binaural impulse responses:

$$\tau_{\text{ext}}(\mathbf{r}) = \mathcal{E}_{\text{ITD}}[a_L(t, \mathbf{r}), a_R(t, \mathbf{r})], \tag{7}$$

where $\mathcal{E}_{\text{ITD}}$ is the Barumerli feature extractor's broadband envelope-correlation estimator, implemented as AMT `itdestimator` with the `MaxIACCe` convention. For the reported methods, reference and reconstructed HRTFs define separate delay fields,

$$\tau(\mathbf{r}) = \tau_{\text{ext}}(\mathbf{r}), \qquad \hat{\tau}(\mathbf{r}) = \hat{\tau}_{\text{ext}}(\mathbf{r}), \tag{8}$$

so that timing errors contribute to the Fisher-tensor comparison.

The delay is mapped onto the transformed ITD scale used by the observer model [15], [9]:

$$\begin{aligned} T(\tau) = \frac{\operatorname{sgn}(\tau)}{0.095} \Big[& \ln\big(32.5 \times 10^{-6} + 0.095|\tau|\big) \\ & - \ln\big(32.5 \times 10^{-6}\big) \Big]. \end{aligned} \tag{9}$$

This is the Reijniers et al. variance-stabilising transformation for heteroscedastic ITD discrimination: empirical broadband ITD JNDs increase approximately linearly with $|\tau|$, and $T(\tau)$ is the corresponding accumulated-JND coordinate. A constant Gaussian noise term on this transformed coordinate therefore represents constant uncertainty in ITD JND units, not constant uncertainty in seconds. The ITD cue used in the spatial cue vector is therefore

$$\text{ITD}(\mathbf{r}) = T(\tau(\mathbf{r})), \qquad \widehat{\text{ITD}}(\mathbf{r}) = T(\hat{\tau}(\mathbf{r})). \tag{10}$$

III. Continuous Spatial Gradients and Tangent Projection

Let the monaural spectral feature vector collect the $B - d$ PGE values for both ears,

$$\mathbf{s}(\mathbf{r}) \in \mathbb{R}^{N_S}, \qquad N_S = 2(B - d).$$

The full spatial cue vector is

$$\mathbf{q}(\mathbf{r}) = \big[\mathbf{s}(\mathbf{r})^T, \text{ILD}(\mathbf{r}), \text{ITD}(\mathbf{r})\big]^T \in \mathbb{R}^M, \qquad M = N_S + 2.$$

For a reconstructed HRTF field, the same feature extraction gives $\widehat{\mathbf{q}}(\mathbf{r})$ on the same spatial grid.

A. Spherical Harmonic Representation

Each scalar cue component is represented on the sphere using a truncated real spherical-harmonic (SH) basis of maximum degree $N_{\max}$. Let

$$\mathbf{Y}(\mathbf{r}) \in \mathbb{R}^C, \qquad C = (N_{\max} + 1)^2,$$

be the vector of basis functions evaluated at $\mathbf{r}$. Cue samples at fitting directions $\{\mathbf{r}_n\}_{n=1}^{N_{\text{fit}}}$ are collected in

$$\mathbf{Q} = [\mathbf{q}(\mathbf{r}_1), \ldots, \mathbf{q}(\mathbf{r}_{N_{\text{fit}}})]^T \in \mathbb{R}^{N_{\text{fit}} \times M},$$

and the SH design matrix is defined as

$$\mathbf{Y}_{\text{fit}} = [\mathbf{Y}(\mathbf{r}_1), \ldots, \mathbf{Y}(\mathbf{r}_{N_{\text{fit}}})]^T \in \mathbb{R}^{N_{\text{fit}} \times C}.$$

The SH coefficient matrix $\mathbf{B} \in \mathbb{R}^{C \times M}$ is obtained by least squares:

$$\mathbf{B} = \mathbf{Y}_{\text{fit}}^{\dagger} \mathbf{Q}, \tag{11}$$

where $\mathbf{Y}_{\text{fit}}^{\dagger}$ denotes the Moore–Penrose pseudoinverse. When $\mathbf{Y}_{\text{fit}}$ has full column rank,

$$\mathbf{Y}_{\text{fit}}^{\dagger} = \left(\mathbf{Y}_{\text{fit}}^T \mathbf{Y}_{\text{fit}}\right)^{-1} \mathbf{Y}_{\text{fit}}^T. \tag{12}$$

The smooth cue field is

$$\widetilde{\mathbf{q}}(\mathbf{r}) = \mathbf{B}^T \mathbf{Y}(\mathbf{r}). \tag{13}$$

For the reconstructed HRTF field, define

$$\widehat{\mathbf{Q}} = [\widehat{\mathbf{q}}(\mathbf{r}_1), \ldots, \widehat{\mathbf{q}}(\mathbf{r}_{N_{\text{fit}}})]^T.$$

It is fitted with the same $N_{\max}$, basis convention, and fitting grid:

$$\widehat{\mathbf{B}} = \mathbf{Y}_{\text{fit}}^{\dagger} \widehat{\mathbf{Q}}, \qquad \widehat{\widetilde{\mathbf{q}}}(\mathbf{r}) = \widehat{\mathbf{B}}^T \mathbf{Y}(\mathbf{r}). \tag{14}$$

Fixing this smoothing operation is necessary because its bandwidth affects the measured spatial gradients.

B. Continuous Spatial Derivatives

The SH representation provides a smooth approximation to each cue field, allowing spatial derivatives to be evaluated at arbitrary directions. Let

$$\mathbf{J}_{\mathbf{Y}}(\mathbf{r}) = \nabla_{\mathbf{r}} \mathbf{Y}(\mathbf{r}) \in \mathbb{R}^{C \times 3}.$$

The Cartesian cue Jacobian is then

$$\mathbf{J}_{\text{Cart}}(\mathbf{r}) = \nabla_{\mathbf{r}} \widetilde{\mathbf{q}}(\mathbf{r}) = \mathbf{B}^T \mathbf{J}_{\mathbf{Y}}(\mathbf{r}). \tag{15}$$

C. Local Tangent Plane Projection

Because source direction lies on the unit sphere, only tangential variations are relevant for local discriminability. The unit normal at $\mathbf{r}$ is $\mathbf{r}$ itself. A local orthonormal tangent basis $\mathbf{V}(\mathbf{r}) = [\mathbf{u}_1, \mathbf{u}_2] \in \mathbb{R}^{3 \times 2}$ is defined by selecting an auxiliary vector

$$\mathbf{h}_{\text{aux}} = \begin{cases} [0, 1, 0]^T, & \text{if } |r_z| > 0.9, \\ [0, 0, 1]^T, & \text{otherwise}, \end{cases}$$

and then setting

$$\mathbf{u}_1 = \frac{\mathbf{h}_{\text{aux}} \times \mathbf{r}}{\|\mathbf{h}_{\text{aux}} \times \mathbf{r}\|_2}, \qquad \mathbf{u}_2 = \mathbf{r} \times \mathbf{u}_1. \tag{16}$$

The same basis is used for target and reconstruction at each direction. Since $\mathbf{r}$ is unit length, its tangent coordinates describe infinitesimal angular displacement in radians. The tangent Jacobian is

$$\mathbf{J}_{\tan}(\mathbf{r}) = \mathbf{J}_{\text{Cart}}(\mathbf{r})\, \mathbf{V}(\mathbf{r}) \in \mathbb{R}^{M \times 2}. \tag{17}$$

IV. Construction of the Fisher–Rao Metric Tensor

For the Fisher analysis, cue observations are modelled locally as Gaussian with mean $\widetilde{\mathbf{q}}(\mathbf{r})$ and direction-independent covariance $\boldsymbol{\Sigma}$. Following the independent-feature covariance used by Barumerli et al. [9], cue-family noise is represented by the block-diagonal matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{\text{mon}}^2 \mathbf{I}_{N_S} & \mathbf{0} & \mathbf{0} \\ \mathbf{0}^T & \sigma_{\text{ILD}}^2 & 0 \\ \mathbf{0}^T & 0 & \sigma_{\text{ITD}}^2 \end{bmatrix}, \tag{18}$$

where $\mathbf{I}_{N_S}$ is the $N_S \times N_S$ identity matrix. The off-diagonal entries of $\mathbf{g}(\mathbf{r})$ are the precision-weighted inner products between tangent cue gradients.

The standard deviations set the relative cue weights. The analysis uses $\sigma_{\text{mon}} = 1.25\,\text{dB}$, $\sigma_{\text{ILD}} = 1\,\text{dB}$, and $\sigma_{\text{ITD}} = 0.569$ on the transformed $T(\tau)$ scale. The ITD value is the Reijniers et al. discrimination-derived standard deviation on the transformed JND coordinate [15]. The monaural and ILD terms are retained as fixed Barumerli-family observer weights for the PGE and broadband-ILD feature convention [9]. They define a reproducible generic observer for comparing reconstructions; they are not refitted here to behavioural thresholds. The Bayesian evaluation additionally includes the observer prior, decision rule, and response stage.

For the diagonal Gaussian covariance defined by these standard deviations, the Fisher information matrix on the two-dimensional tangent space is

$$\mathbf{g}(\mathbf{r}) = \mathbf{J}_{\tan}(\mathbf{r})^T \, \boldsymbol{\Sigma}^{-1} \, \mathbf{J}_{\tan}(\mathbf{r}) \in \mathbb{R}^{2\times 2}. \tag{19}$$

Let $\mathbf{j}_u(\mathbf{r}) = \nabla_{\tan} u(\mathbf{r}) \in \mathbb{R}^2$ denote the tangent gradient of a scalar cue u. The FIM can be written explicitly as

$$\begin{aligned} \mathbf{g}(\mathbf{r}) = \frac{1}{\sigma_{\text{mon}}^2} \sum_{c\in\{L,R\}} \sum_{b=1}^{B-d} \mathbf{j}_{S_c[b]} \mathbf{j}_{S_c[b]}^T \\ + \frac{1}{\sigma_{\text{ILD}}^2} \mathbf{j}_{\text{ILD}} \mathbf{j}_{\text{ILD}}^T + \frac{1}{\sigma_{\text{ITD}}^2} \mathbf{j}_{\text{ITD}} \mathbf{j}_{\text{ITD}}^T, \end{aligned} \tag{20}$$

where every cue gradient on the right-hand side is evaluated at $\mathbf{r}$. Thus, $\mathbf{g}$ is positive semidefinite and is positive definite only where the weighted cue gradients span both tangent directions. Strict positive definiteness for tensor comparison is supplied by the regularisation defined below.

Figures 1 and 2 place the two observer summaries side by side: the CRB ellipses retain the local tensor anisotropy, whereas the Barumerli MAP response-error distributions show the more radially symmetric covariance induced by the response-scatter stage.

V. AIRM Tensor Discrepancy

The target and reconstructed Fisher tensors are compared using the affine-invariant Riemannian metric (AIRM) on symmetric positive-definite matrices. Since empirical Fisher tensors may be singular where the weighted cue gradients do not span both tangent directions, both fields are symmetrised and regularised by the same diagonal term,

$$\mathbf{g}_\eta(\mathbf{r}) = \mathbf{g}(\mathbf{r}) + \eta \mathbf{I}_2, \qquad \widehat{\mathbf{g}}_\eta(\mathbf{r}) = \widehat{\mathbf{g}}(\mathbf{r}) + \eta \mathbf{I}_2, \tag{21}$$

with $\eta = 10^{-6}$. The pointwise tensor discrepancy is

$$d_{\text{AIRM}}(\mathbf{g}_\eta, \widehat{\mathbf{g}}_\eta) = \left\| \log\left(\mathbf{g}_\eta^{-1/2} \widehat{\mathbf{g}}_\eta \mathbf{g}_\eta^{-1/2} \right) \right\|_F. \tag{22}$$

For a 2×2 tensor pair this is evaluated from the positive generalised eigenvalues λ_1, λ_2 of $(\widehat{\mathbf{g}}_\eta, \mathbf{g}_\eta)$:

$$d_{\text{AIRM}} = \sqrt{\log^2(\lambda_1) + \log^2(\lambda_2)}. \tag{23}$$

Numerically, the same result is obtained by Cholesky congruence: if $\mathbf{g}_\eta = \mathbf{L}\mathbf{L}^T$, the eigenvalues are computed from the symmetrised matrix $\mathbf{L}^{-1} \widehat{\mathbf{g}}_\eta \mathbf{L}^{-T}$, with eigenvalues below machine precision floored before taking logarithms. The mean AIRM discrepancy over the evaluation grid $\{\mathbf{r}_n\}_{n=1}^{N_{\text{eval}}}$ is

$$\mathcal{E}_{\text{AIRM}} = \frac{1}{N_{\text{eval}}} \sum_{n=1}^{N_{\text{eval}}} d_{\text{AIRM}}\big(\mathbf{g}_\eta(\mathbf{r}_n), \widehat{\mathbf{g}}_\eta(\mathbf{r}_n)\big). \tag{24}$$

VI. Evaluation Metrics

AIRM measures the distance between reference and reconstructed Fisher tensors. Signal-domain metrics measure physical reconstruction error, and the Bayesian-observer metrics summarise simulated localisation responses.

A. Log-Spectral Distortion (LSD)

Using the HRTF-magnitude notation defined above, the binaural log-spectral distortion is

$$\begin{aligned} \text{LSD} = \frac{1}{N_{\text{eval}}} \sum_{n=1}^{N_{\text{eval}}} \Bigg[\frac{1}{2W} \sum_{c\in\{L,R\}} \sum_{w=1}^{W} \\ \left(20 \log_{10} \frac{A_c(f_w, \mathbf{r}_n) + \varepsilon_A}{\widehat{A}_c(f_w, \mathbf{r}_n) + \varepsilon_A} \right)^2 \Bigg]^{1/2}. \end{aligned} \tag{25}$$

LSD serves as the principal signal-domain metric because it directly measures spectral magnitude agreement. For the HRIR/SOFA reconstructions evaluated here, LSD is computed over the common LAP/SAM band from 20 Hz to 20 kHz, except for FSP-AE, for which the upper limit is 16 kHz to match the model's configured spectral support.

B. Interaural Level Difference Error

ILD error measures binaural level reconstruction. Define the frequency-averaged HRTF ILD

$$\text{ILD}_{\text{HRTF}}(\mathbf{r}) = \frac{1}{F} \sum_{k=1}^{F} 20 \log_{10} \left(\frac{A_L(f_k, \mathbf{r}) + \varepsilon_A}{A_R(f_k, \mathbf{r}) + \varepsilon_A} \right), \tag{26}$$

with reconstruction error

$$\mathcal{E}_{\text{ILD}} = \frac{1}{N_{\text{eval}}} \sum_{n=1}^{N_{\text{eval}}} \left| \text{ILD}_{\text{HRTF}}(\mathbf{r}_n) - \widehat{\text{ILD}}_{\text{HRTF}}(\mathbf{r}_n) \right|. \tag{27}$$

This signal-domain quantity differs from the broadband DTF power-ratio ILD cue used in the Fisher tensor. One measures error in the binaural level field; the other contributes its spatial derivatives to the Fisher matrix.

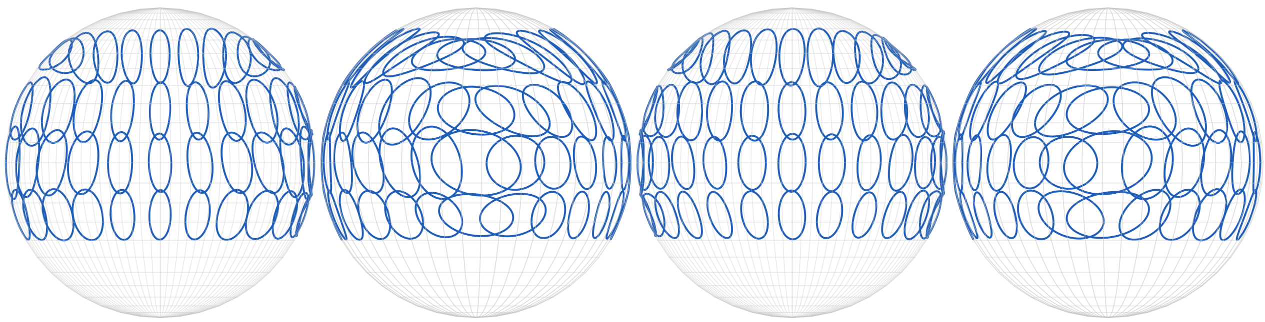

Fig. 1. Local auditory discrimination ellipses derived from the Cramér–Rao bound. Ellipses show the 95% two-dimensional covariance contour, scaled by $\sqrt{\chi^2_{2,0.95}}$. Views correspond to azimuths of $0°$, $90°$, $180°$, and $270°$. Results are based on the median Fisher tensor of the 41-subject SONICOM evaluation cohort.

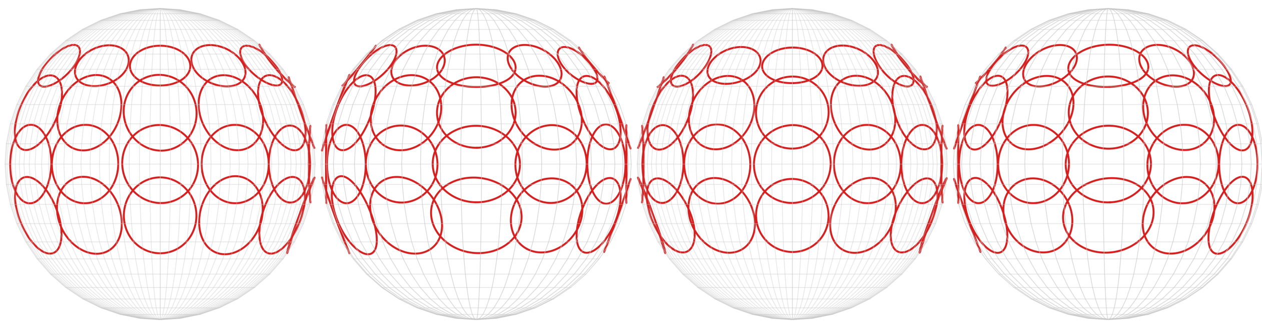

Fig. 2. Predicted pointing error from a converged Monte Carlo simulation of the maximum a posteriori estimator of Barumerli et al. [9]. Ellipses represent local response covariance matrices, scaled to one standard deviation along each principal axis. Views correspond to azimuths of $0°$, $90°$, $180°$, and $270°$. Results are based on the median covariance of the 41-subject SONICOM evaluation cohort.

C. Fisher-Tensor AIRM Discrepancy

The primary tensor metric is the mean AIRM discrepancy between regularised target and reconstructed Fisher tensors:

$$\mathcal{E}_{\mathrm{AIRM}} = \frac{1}{N_{\mathrm{eval}}} \sum_{n=1}^{N_{\mathrm{eval}}} d_{\mathrm{AIRM}}\big(\mathbf{g}_\eta(\mathbf{r}_n), \widehat{\mathbf{g}}_\eta(\mathbf{r}_n)\big). \tag{28}$$

This metric compares the anisotropy, orientation, and scale of the Fisher tensors formed from the spatial cue gradients.

The scalar AIRM score is supplemented by tensor summaries. Let $\mu_1(\mathbf{r}) \geq \mu_2(\mathbf{r}) > 0$ be the eigenvalues of $\mathbf{g}_\eta(\mathbf{r})$, and let $\mathbf{v}_1(\mathbf{r})$ be its principal information direction. The reported summaries are

$$\mathcal{E}_{\mathrm{det}} = \frac{1}{N_{\mathrm{eval}}} \sum_n \left| \log \frac{\det \widehat{\mathbf{g}}_\eta(\mathbf{r}_n)}{\det \mathbf{g}_\eta(\mathbf{r}_n)} \right|, \tag{29}$$

$$\mathcal{E}_{\mathrm{anis}} = \frac{1}{N_{\mathrm{eval}}} \sum_n \left| \log \frac{\hat{\mu}_1(\mathbf{r}_n)/\hat{\mu}_2(\mathbf{r}_n)}{\mu_1(\mathbf{r}_n)/\mu_2(\mathbf{r}_n)} \right|, \tag{30}$$

$$\mathcal{E}_{\mathrm{orient}} = \frac{1}{|\mathcal{R}_{\mathrm{aniso}}|} \sum_{\mathbf{r}_n \in \mathcal{R}_{\mathrm{aniso}}} \arccos\left(\left| \widehat{\mathbf{v}}_1(\mathbf{r}_n)^T \mathbf{v}_1(\mathbf{r}_n) \right| \right), \tag{31}$$

where $\mathcal{R}_{\mathrm{aniso}}$ contains the directions for which both tensors satisfy $\mu_1/\mu_2 \geq 1.10$. The orientation error is reported in degrees. The determinant, anisotropy, and orientation summaries decompose the changes measured jointly by AIRM.

D. Bayesian Localisation Metrics

A Bayesian localisation model supplies lateral RMS error, local polar RMS error, and quadrant error [19], [9]. Let ℓ_i, φ_i denote the lateral and polar target angles and $\widetilde{\ell}_i, \widetilde{\varphi}_i$ their simulated responses for N_{trial} responses, where N_{trial} includes the simulated repetitions over the dense target-direction set. For the polar-error target set $\mathcal{P}$, define local polar responses by

$$\mathcal{A} = \{ i \in \mathcal{P} : |\mathrm{wrap}(\widetilde{\varphi}_i - \varphi_i)| < 90° \}.$$

The reported metrics are

$$\text{Lateral RMS} = \sqrt{\frac{1}{N_{\mathrm{trial}}} \sum_{i=1}^{N_{\mathrm{trial}}} (\widetilde{\ell}_i - \ell_i)^2}, \tag{32}$$

$$\text{Local Polar RMS} = \sqrt{\frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \mathrm{wrap}(\widetilde{\varphi}_i - \varphi_i)^2}, \tag{33}$$

$$\text{Quadrant Error} = \left(1 - \frac{|\mathcal{A}|}{|\mathcal{P}|} \right) \times 100. \tag{34}$$

The implementation uses AMT `barumerli2023` with DTF feature extraction, maximum a posteriori estimation, and $R = 300$ simulated repetitions per target direction. Its parameters are $\sigma_{\mathrm{ITD}} = 0.569$, $\sigma_{\mathrm{ILD}} = 1$, $\sigma_{\mathrm{spectral}} = 4$, $\sigma_{\mathrm{prior}} = 11.5$, and $\sigma_{\mathrm{motor}} = 14$. The resulting metrics summarise simulated pointing after application of the complete observer model.

VII. Evaluation Procedure

A. Data and Spatial Conditions

The evaluation uses the SONICOM HRTF dataset [10] with a 162/41 train–test split. The retained-direction counts

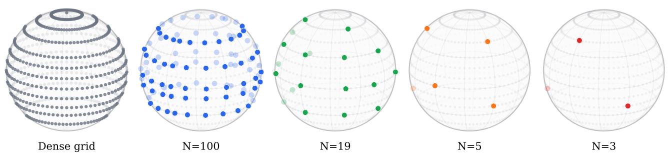

Fig. 3. Dense SONICOM evaluation grid and retained sparse sampling schemes used for the four upsampling conditions. Grey points denote the dense 793-direction target grid; coloured points denote retained directions.

and sparse masks follow the SONICOM upsampling protocol used by Hu et al. for HRTFformer, within the broader LAP upsampling benchmark context [11], [12]. RANF, FSP-AE, and the interpolation methods use the same 41 held-out subjects and sparse masks. The test identifiers and retained coordinates are archived with the evaluation outputs. The magnitude representation has 128 positive-frequency bins per ear after removal of the DC bin, giving 256 binaural magnitude values per direction on a dense grid of 793 directions.

Four sparse-to-dense conditions are evaluated: $100 \to 793$, $19 \to 793$, $5 \to 793$, and $3 \to 793$. The same retained directions are used for every method and subject within each condition (Fig. 3).

All Fisher-tensor fitting and AIRM comparison are performed on the unique physical grid. Duplicate coordinate entries introduced by an implementation layout receive no additional weight.

B. Compared Reconstruction Methods

The evaluation requires a reconstructed HRTF field on the common evaluation grid. The SUpDEq configurations follow the directional-equalization approach of Pörschmann et al. [13]. The reported comparison includes:

1) **SUpDEq-SH:** SUpDEq preprocessing followed by SH interpolation.
2) **SUpDEq-MCA:** the published Magnitude-Corrected and Time-Aligned configuration, which uses the same SH interpolation stage and then applies magnitude correction.
3) **Natural- and barycentric-MCA variants:** explicitly labelled $6\,\mathrm{dB}$-limited SUpDEq variants using natural-neighbour and barycentric interpolation, respectively.
4) **RANF:** a retrieval-augmented neural field that predicts magnitude and ITD and reconstructs HRIRs using a minimum-phase representation with ITD compensation [16], [17].
5) **FSP-AE:** a neural network conditioned on source position and frequency that predicts HRTF magnitude and ITD before HRIR reconstruction [18].

The natural-neighbour variant uses Voronoi-weighted interpolation as implemented by the SFS/SUpDEq routines. SUpDEq-Bary-MCA is undefined at three retained directions because they cannot form the required closed spherical hull; this condition is marked not applicable.

For SUpDEq calls using SH interpolation, the reconstruction order is capped at $\min(6, \lfloor\sqrt{N_{\mathrm{ret}}} - 1\rfloor)$, where N_{ret} is the number of retained directions. This avoids fitting more SH coefficients than the sparse grid can support. At $N_{\mathrm{ret}} = 3$, the cap therefore gives a zeroth-order SH magnitude field; methods without a direction-dependent delay or correction stage can consequently yield nearly constant cue fields. The Fisher-tensor metric records this behaviour as interpolation failure.

C. Preprocessing and Reconstruction Normalisation

All methods are converted to a common representation before comparison. Left- and right-ear HRTF magnitudes are aligned on the same frequency bins, with the DC bin removed. Methods operating in the SH domain are inverse-transformed to the full spatial grid; methods outputting HRIRs are converted to magnitudes by a common procedure. Fisher cues are extracted using the Barumerli PGE, ILD, and `MaxIACCe` ITD convention. RANF and FSP-AE outputs are treated as HRIR/SOFA reconstructions with model-provided ITD.

The same CTF removal is applied to target and reconstruction before DTF feature extraction, leaving the direction-dependent spectral variation used to form the Fisher tensors.

VIII. Results

A. Run Completion and Applicability

The reported comparison comprises 984 method–condition–subject rows: 41 subjects, four retention conditions, and six reconstruction methods. Of these rows, 943 completed successfully. The remaining 41 rows were all the $N = 3$ SUpDEq-Bary-MCA condition, which is not applicable because three retained directions cannot define the closed spherical hull required by the barycentric interpolation step. No completed rows contain missing AIRM, LSD, ILD, or Bayesian localisation metrics. All completed Fisher tensors use the same Barumerli PGE, ILD, and `MaxIACCe` ITD cue convention, and all reported Bayesian-observer summaries use 300 simulated repetitions.

B. Fisher-Tensor Results

Figure 4 reports cohort-mean AIRM for the six reconstruction methods. RANF gives the lowest mean AIRM at every retention condition, ranging from 0.48 at $N = 100$ to 0.68 at $N = 3$. FSP-AE is close to RANF at $N = 100$ and is second-best at $N = 19$, $N = 5$, and $N = 3$. Among the interpolation methods, SUpDEq-Natural-MCA and SUpDEq-Bary-MCA are strongest at $N = 100$ and $N = 19$, but SUpDEq-Natural-MCA produces very large AIRM discrepancies at $N = 5$ and $N = 3$. SUpDEq-Bary-MCA is not applicable at $N = 3$.

AIRM depends on local derivatives and on tensor anisotropy and orientation, so its method ranking can differ from rankings based on LSD or Bayesian-observer summaries.

Table I decomposes the AIRM behaviour into changes in tensor scale, eccentricity, and principal-axis direction. It reports cohort means; orientation error is evaluated only where both reference and reconstructed tensors are sufficiently anisotropic for a principal-axis angle to be meaningful.

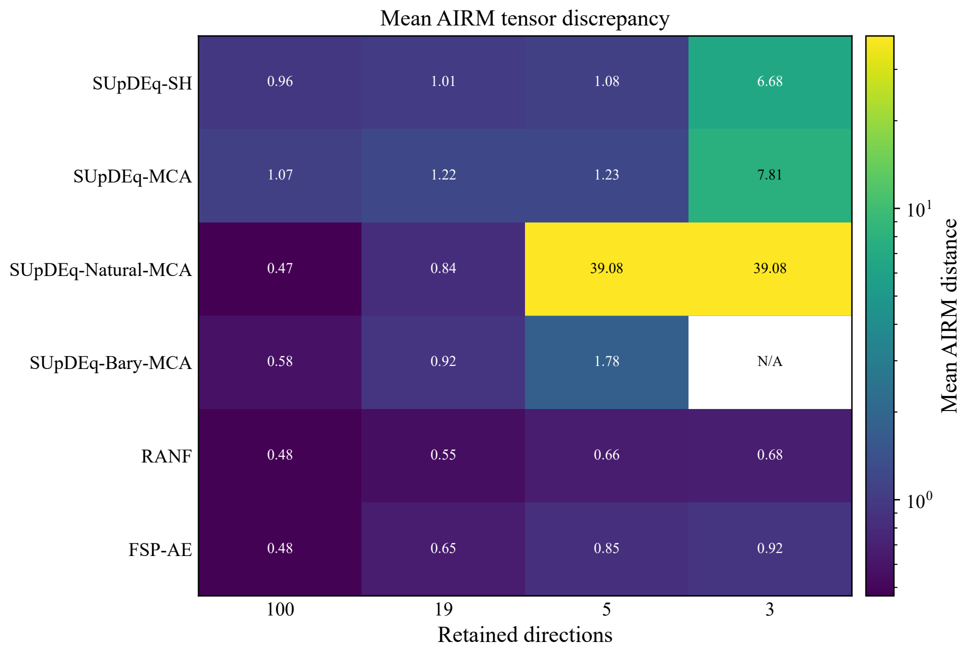

Fig. 4. Mean Fisher-tensor AIRM discrepancy over the 41-subject cohort. Lower values indicate closer preservation of the dense-reference tensor field; the blank entry is the not-applicable $N = 3$ SUpDEq-Bary-MCA condition.

TABLE I
MEAN SECONDARY FISHER-TENSOR COMPONENT ERRORS OVER THE 41-SUBJECT COHORT FOR THE PRINCIPAL COMPARISON SET. DASHES DENOTE CONDITIONS THAT ARE NOT APPLICABLE OR FOR WHICH THE COMPONENT SUMMARY IS UNDEFINED UNDER THE STATED VALIDITY CRITERIA.

Component	Method	$N = 100$	$N = 19$	$N = 5$	$N = 3$
Log-det.	SUpDEq-SH	0.70	0.68	0.87	7.14
Log-det.	SUpDEq-MCA	0.90	0.95	0.97	8.70
Log-det.	SUpDEq-Natural-MCA	0.32	0.69	–	–
Log-det.	SUpDEq-Bary-MCA	0.37	0.67	1.31	–
Log-det.	RANF	0.35	0.39	0.48	0.50
Log-det.	FSP-AE	0.36	0.57	0.81	0.92
Anisotropy	SUpDEq-SH	0.56	0.67	0.67	5.80
Anisotropy	SUpDEq-MCA	0.57	0.68	0.80	6.41
Anisotropy	SUpDEq-Natural-MCA	0.31	0.54	–	–
Anisotropy	SUpDEq-Bary-MCA	0.38	0.54	1.43	–
Anisotropy	RANF	0.33	0.37	0.44	0.45
Anisotropy	FSP-AE	0.33	0.43	0.55	0.60
Orientation ($^\circ$)	SUpDEq-SH	18.9	20.2	16.4	13.0
Orientation ($^\circ$)	SUpDEq-MCA	19.3	22.5	17.1	13.2
Orientation ($^\circ$)	SUpDEq-Natural-MCA	7.2	11.3	–	–
Orientation ($^\circ$)	SUpDEq-Bary-MCA	8.6	14.3	17.7	–
Orientation ($^\circ$)	RANF	7.4	9.0	10.7	10.6
Orientation ($^\circ$)	FSP-AE	7.6	8.7	9.6	9.6

C. Signal and Bayesian-Observer Metrics

Conventional and Bayesian-observer metrics were computed for every completed method–condition–subject row. The Bayesian localisation model produces absolute simulated errors, but the manuscript reports the boxplot summaries relative to the dense self-reference condition for the same subject and observer configuration. These relative values isolate the additional model error introduced by the reconstruction, since the generic Barumerli listener already has non-zero error when the measured HRTF is compared with itself.

RANF has the lowest LSD throughout the evaluated retention conditions. FSP-AE has larger LSD than RANF, but is competitive with the interpolation baselines and remains spectrally stable as sparsity increases. It also avoids the severe Fisher-tensor degradation observed for several interpolation baselines. Among the interpolation methods, AIRM, LSD, and relative Bayesian-observer summaries do not always select the same reconstruction. The three evaluations measure tensor preservation, spectral reconstruction, and simulated localisation, respectively.

Figure 6 reports subject-wise distributions for lateral RMS, local polar RMS, and quadrant error. Lateral RMS captures horizontal response spread; local polar RMS and quadrant error capture elevation error and front–back confusions.

D. Metric Relationships

Figure 7 compares AIRM with conventional and Bayesian-observer summaries across the 943 completed rows. Because the rows form a repeated subject–method–condition design, inferential p-values are not computed by treating the rows

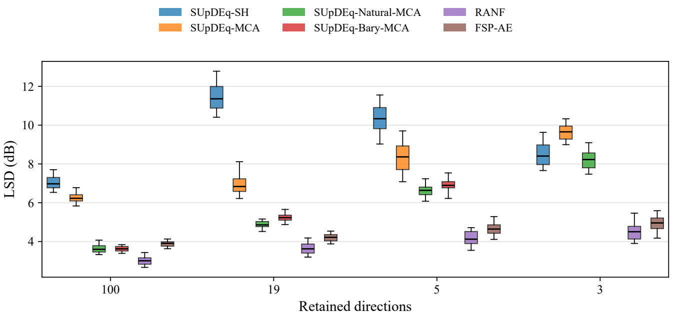

Fig. 5. Subject-wise LSD distributions over the 41 held-out subjects. Boxes show medians and interquartile ranges; whiskers span the 5th–95th percentiles.

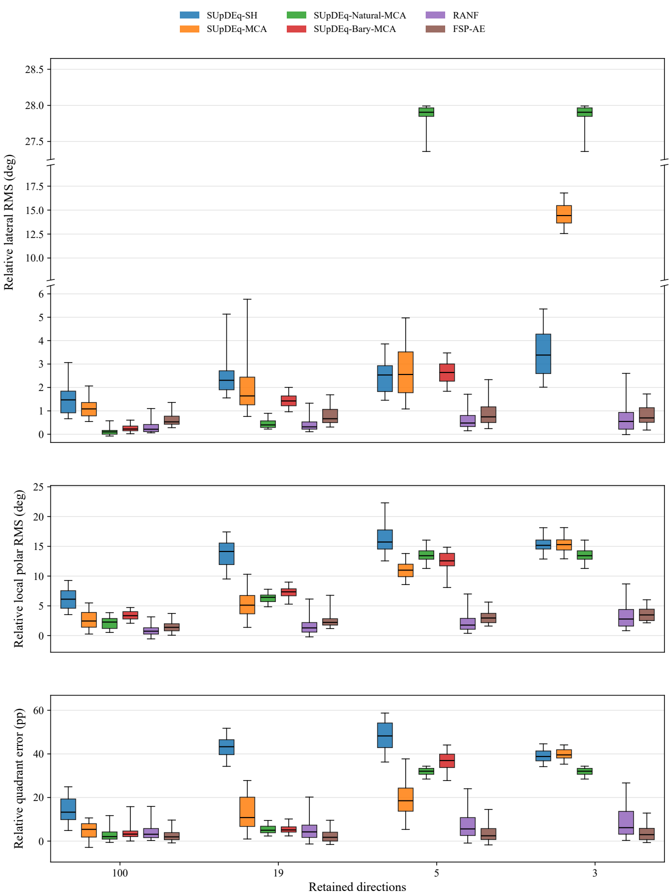

Fig. 6. Subject-wise Bayesian-observer degradation over the 41 held-out subjects. Boxes show medians and interquartile ranges; whiskers span the 5th–95th percentiles. Metrics are relative to the dense self-reference condition, and the lateral RMS panel uses a broken vertical axis to show compact and high-error distributions together.

as independent samples. Table II instead reports correlations over the 23 completed method–retention means, together with subject-cluster bootstrap intervals for the corresponding row-wise Spearman coefficients. AIRM and LSD have a weak linear association but a strong rank association, indicating that the ordering of reconstruction quality is substantially shared even though the absolute scales are not linearly equivalent. AIRM is also strongly rank-associated with the Bayesian-observer summaries. These relationships are approximately monotonic, but they are not one-to-one. Several methods with similar LSD differ appreciably in AIRM, and some methods with comparable Bayesian localisation summaries differ in

TABLE II

ASSOCIATION BETWEEN MEAN AIRM AND ESTABLISHED EVALUATION METRICS. CORRELATIONS ARE COMPUTED OVER THE 23 COMPLETED METHOD–RETENTION MEANS; THE FINAL COLUMN GIVES A 95% SUBJECT-CLUSTER BOOTSTRAP INTERVAL FOR THE CORRESPONDING ROW-WISE SPEARMAN COEFFICIENT.

Metric	Pearson r	Spearman ρ	Row-wise ρ 95% CI
LSD	0.26	0.86	[0.82, 0.85]
ILD	0.41	0.81	[0.77, 0.80]
Relative lateral RMS	0.97	0.95	[0.85, 0.89]
Relative local polar RMS	0.48	0.85	[0.75, 0.80]
Relative quadrant error	0.39	0.78	[0.65, 0.72]

TABLE III

POPULATION-MEDIAN FISHER-PREDICTED LOCAL $d' = 1$ ANGULAR THRESHOLDS IN DEGREES. LATERAL COLUMNS USE HORIZONTAL-PLANE AZIMUTHAL DISPLACEMENTS; VERTICAL COLUMNS USE FRONTAL-PLANE ELEVATIONAL DISPLACEMENTS.

Method	N	Lat. $(0,0)$	Lat. $(45,0)$	Lat. $(90,0)$	Vert. $(0,0)$	Vert. $(0,45)$	Err.
Measured	–	1.70	3.08	6.78	4.44	5.71	0.00
SUpDEq-MCA	19	1.84	2.55	5.66	4.27	5.00	0.54
SUpDEq-MCA	5	1.66	2.62	5.45	6.83	5.93	0.89
SUpDEq-Bary-MCA	19	1.83	3.02	5.77	6.31	7.04	0.88
SUpDEq-Bary-MCA	5	2.11	3.08	6.38	5.63	6.39	0.53
RANF	19	1.71	2.88	6.80	4.45	6.36	0.18
RANF	5	1.85	2.90	7.76	4.48	7.23	0.57
FSP-AE	19	1.81	3.32	7.76	4.72	6.53	0.49
FSP-AE	5	1.74	3.31	9.44	6.52	9.59	1.78

tensor discrepancy.

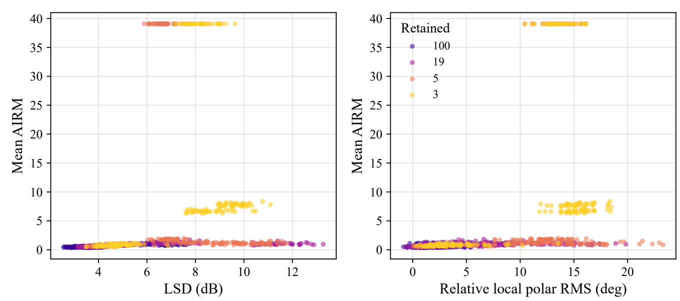

Fig. 7. Mean AIRM discrepancy against LSD and relative local polar RMS across completed subject–method–condition rows. Correlation summaries are reported in Table II.

E. Local Discrimination-Threshold Plausibility

AIRM is a tensor distance, not itself an angular threshold. To express selected Fisher tensors in a more directly psychophysical unit, the $d' = 1$ threshold defined above was evaluated along lateral unit tangent directions at $(0°, 0°)$, $(45°, 0°)$, and $(90°, 0°)$, and along vertical unit tangent directions at $(0°, 0°)$ and $(0°, 45°)$. Coordinate pairs denote azimuth and elevation. Values are converted from radians to degrees. Table III reports scalar projections of the discrimination ellipse construction in Fig. 1: model-derived thresholds under the Barumerli/Reijniers cue-noise convention, not fitted behavioural minimum audible angles.

The dense reference predicts increasing lateral thresholds from frontal to lateral directions, from $1.70°$ at $(0°, 0°)$ to $6.78°$ at $(90°, 0°)$. The vertical threshold increases from $4.44°$ at $(0°, 0°)$ to $5.71°$ at $(0°, 45°)$. This ordering is consistent with broadband spatial-resolution data: Stevenson-Hoare's Fig. 14 gives horizontal-plane means, collapsed over left and right, of approximately $1.5°$, $2.7°$, and $7.2°$ at $0°$, $45°$, and

90°, respectively [4], while Grantham et al. report 1.6°, 2.8°, and 6.5° for horizontal, 60° diagonal, and vertical wideband-noise trajectories [5]. RANF preserves the five local thresholds most closely at $N = 19$; at $N = 5$, SUpDEq-Bary-MCA and RANF are closest. This local projection complements AIRM by translating selected tensor discrepancies into angular units.

IX. Discussion and Future Work

Bivariate Fisher information evaluates the spatial derivatives of the selected auditory cues, and AIRM measures reconstruction-induced changes in the scale, anisotropy, and orientation of the resulting tensors. This differs from the Bayesian localisation model, which predicts pointing responses from sensory evidence, a prior, a decision rule, and response variability. The two analyses therefore quantify complementary quantities: simulated pointing and local discrimination.

The high-sparsity results identify limits of the interpolation methods. Three retained directions do not satisfy the closed-hull requirement of SUpDEq-Bary-MCA, and SH-based or natural-neighbour interpolation can produce severe Fisher-tensor divergence at extreme sparsity even when conventional error summaries remain finite. RANF gives the strongest overall performance, while FSP-AE provides a second learning-based comparator with stable Fisher-tensor behaviour despite higher LSD.

The present Fisher analysis uses direction-independent cue-noise weights, so spatial variation in predicted discriminability arises from HRTF cue gradients rather than spatially varying internal noise. A more general observer model could allow cue-noise covariance to vary with source direction, separating anisotropy from the acoustic cue field from anisotropy due to direction-dependent sensory reliability. This extension is distinct from the equatorial prior in the Barumerli model: the prior biases point estimates towards more probable source regions, whereas direction-dependent cue noise would change the local discrimination threshold itself.

The Fisher tensor defines a direction-dependent metric on the sphere, so HRTF reconstruction can be viewed as preserving a local derivative structure rather than only matching spectra at sampled directions. A future learning objective could combine spectral error with AIRM as a tensor-valued loss, or more tractably regularise the spatial Jacobians of PGE, ILD, and ITD. The neighbour dissimilarity loss used in HRTFformer is a related step, since it penalises local inconsistency across the reconstructed HRTF field rather than treating each direction as an isolated sample.

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